

PRACTICAL – 3

CODE:

```
bhardwaj@Pranathi: ~/25200
GNU nano 2.7.2 process_states.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;
    pid = fork();
    if (pid < 0)
    {
        printf("Fork Failed!\n");
        return 1;
    }
    if (pid == 0)
    {
        printf("\n===== CHILD PROCESS =====\n");
        printf("Child PID      : %d\n", getpid());
        printf("Parent PID      : %d\n", getppid());
        printf("Child State      : Running\n");
        sleep(10);
        printf("Child State      : Terminated\n");
        exit(0);
    }
    else
    {
        printf("\n===== PARENT PROCESS =====\n");
        printf("Parent PID      : %d\n", getpid());
        printf("Child PID      : %d\n", pid);
        printf("Parent State     : Waiting\n");
        wait(NULL);
        printf("Child Finished.\n");
        printf("Parent State     : Running\n");
        printf("Parent Terminated.\n");
    }
}
```

OUTPUT:

```
bhardwaj@Pranathi: ~/25200
bhardwaj@Pranathi: ~/2520090182_OSSP/2520090182_Practical/Practical-03$ ./process_states
===== PARENT PROCESS =====
Parent PID      : 1766
Child PID       : 1767
Parent State     : Waiting

===== CHILD PROCESS =====
Child PID       : 1767
Parent PID      : 1766
Child State      : Running
Child State      : Terminated
Child Finished.
Parent State     : Running
Parent Terminated.
```

Using ps:

```
bhardwaj@Pranathi: ~/25200
bhardwaj@Pranathi: ~/2520090182_OSSP/2520090182_Practical/Practical-03$ ps -ef | grep process_states
bhardwaj 1776 305 0 06:44 pts/0 00:00:00 grep --color=auto process_states
bhardwaj@Pranathi: ~/2520090182_OSSP/2520090182_Practical/Practical-03$ |
```

Using top:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-03$ top
top - 06:45:26 up 2:49, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.2 id, 0.0 wa, 0.0 hi, 0.8 si, 0.0 st
MiB Mem : 7762.6 total, 6860.7 free, 560.5 used, 498.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7202.1 avail Mem
Locate string |

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21836	13200	9796	S	0.0	0.2	0:00.77	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50364	16708	15604	S	0.0	0.2	0:00.51	systemd-journal
97	root	20	0	25404	6792	5152	S	0.0	0.1	0:01.47	systemd-udev
157	systemd+	20	0	21460	13184	10928	S	0.0	0.2	0:00.12	systemd-resolve
167	systemd+	20	0	91028	7952	6980	S	0.0	0.1	0:00.21	systemd-timesyn
176	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
177	message+	20	0	9592	5384	4652	S	0.0	0.1	0:00.25	dbus-daemon
189	root	20	0	18156	8908	7848	S	0.0	0.1	0:00.13	systemd-logind
192	root	20	0	1756104	14132	10884	S	0.0	0.2	0:00.28	wsl-pro-service
201	syslog	20	0	222516	5860	4636	S	0.0	0.1	0:00.15	rsyslogd
203	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
220	root	20	0	107016	23136	13672	S	0.0	0.3	0:00.08	unattended-upgr
303	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
304	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.13	Relay(305)
305	bhardwaj	20	0	6212	5512	3696	S	0.0	0.1	0:00.12	bash
306	root	20	0	6700	4668	3884	S	0.0	0.1	0:00.01	login

Using proc:

```

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-03$ ls /proc
1 177 283 350 97 cgroups devices fb irq kpagecgroup meminfo net self sysrq-trigger version
1019 1788 220 351 996 cmdline diskstats filesystems kallsyms kpagecount misc pagetypeinfo slabinfo sysvipc vmallocinfo
1209 189 303 378 997 config.gz dma fs kcore kpageflags modules partitions softirqs thread-self vmstat
157 192 304 52 acpi consoles driver interrupts key-users latency_stats mounts pressure stat timer_list zoneinfo
167 2 305 6 buddyinfo cpuinfo dynamic_debug iomem keys loadavg mpt schedstat swaps tty
176 201 306 953 bus crypto execdomains ioports kmsg locks mtrr scsi sys uptime

```